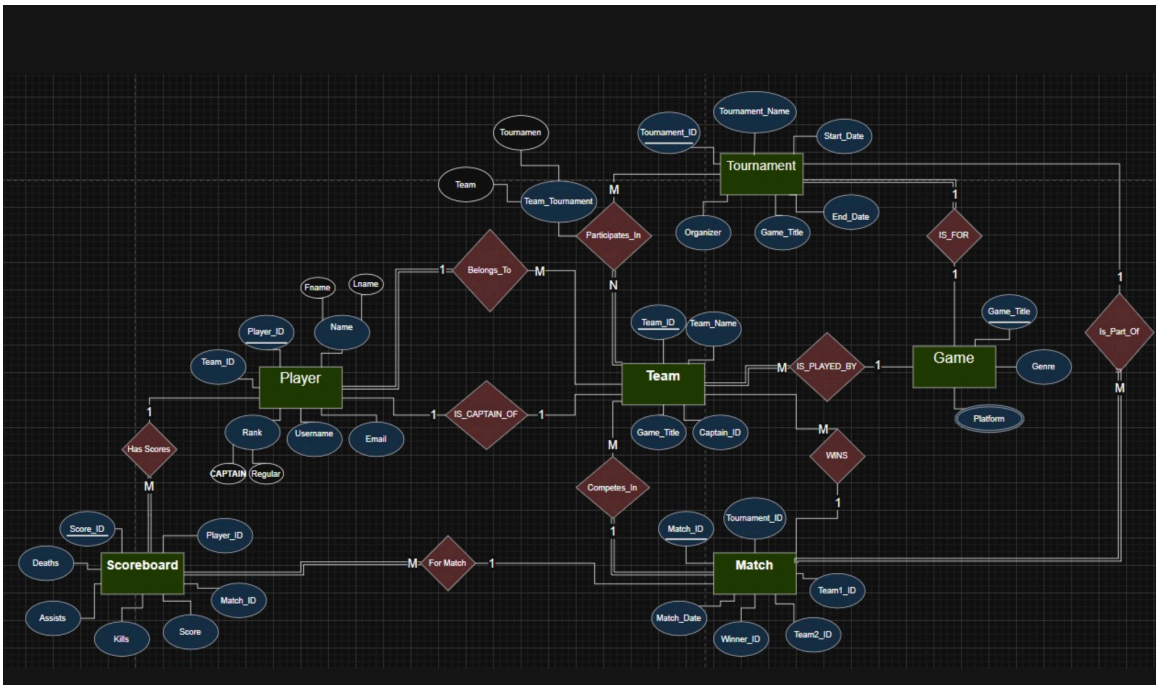
Phase 2 - Normalization & Relational Schema

Game Tournament Management System

1. ER Diagram

The following ER diagram represents the system entities and relationships.



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2. Relational Database Schema (RDB)

PLAYER(Player_ID PK, Name, Username, Email, Rank, Team_ID FK→TEAM.Team_ID)

TEAM(Team_ID PK, Team_Name, Captain_ID FK→PLAYER.Player_ID, Game_Title FK→GAME.Game_Title)

TOURNAMENT(Tournament_ID PK, Tournament_Name, Start_Date, End_Date, Organizer, Game_Title FK→GAME.Game_Title)

MATCH(Match_ID PK, Match_Date, Tournament_ID FK→TOURNAMENT.Tournament_ID, Team1_ID FK→TEAM.Team_ID, Team2_ID FK→TEAM.Team_ID, Winner_ID FK→TEAM.Team_ID)

GAME(Game_Title PK, Genre, Platform)

SCOREBOARD(Score_ID PK, Player_ID FK→PLAYER.Player_ID, Match_ID FK→MATCH.Match_ID,
Score, Kills, Assists, Deaths)

TEAM_TOURNAMENT(Team_ID FK→TEAM.Team_ID, Tournament_ID FK→TOURNAMENT.Tournament_ID,
PK(Team_ID, Tournament_ID))

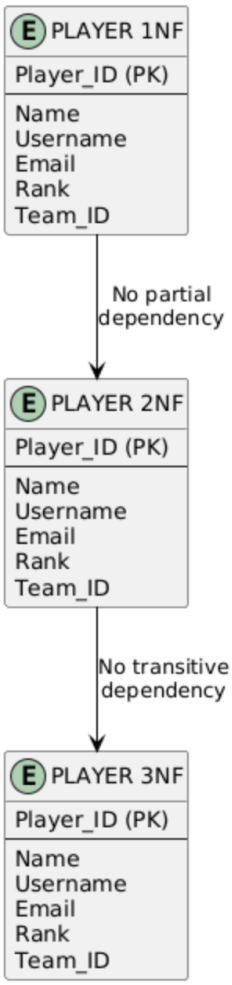
3. Normalization (1NF, 2NF, 3NF)

All relations were analyzed to confirm compliance with 1NF, 2NF and 3NF.

PLAYER

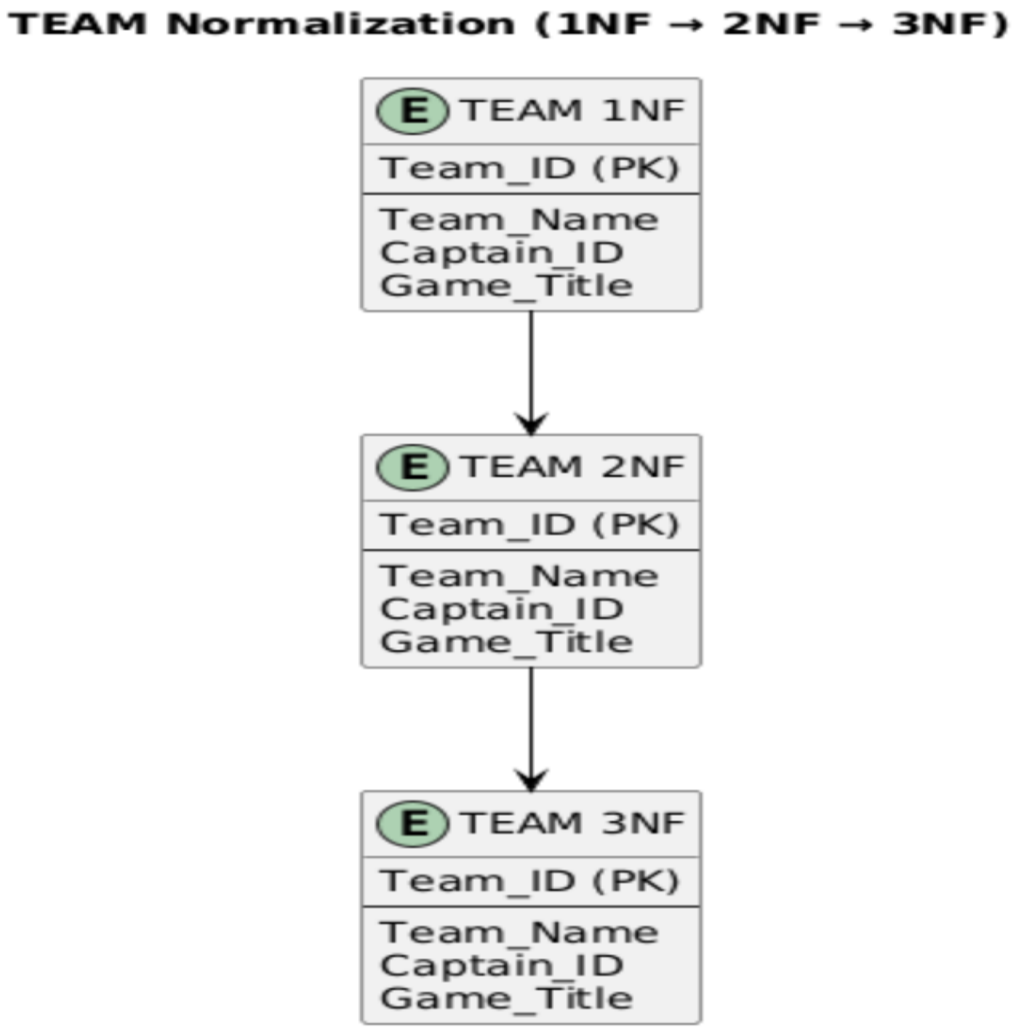
1NF: Atomic.
2NF: PK simple.
3NF: No transitive dependency.

PLAYER Normalization (1NF → 2NF → 3NF)



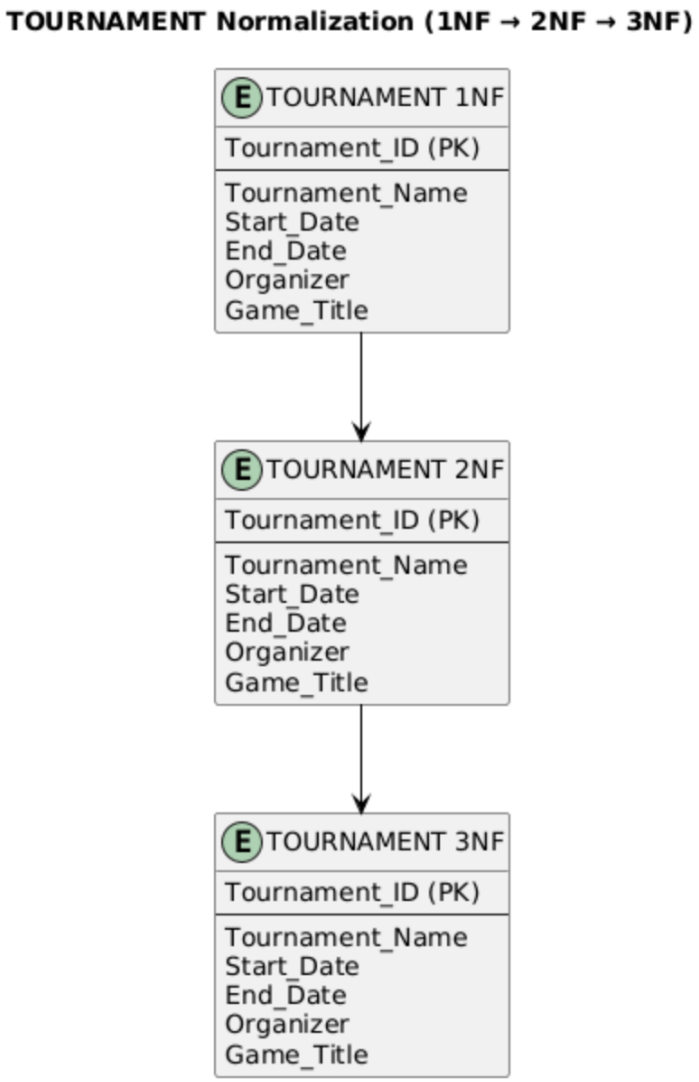
TEAM

1NF: Atomic.
2NF: PK simple.
3NF: No transitive dependency.



TOURNAMENT

1NF: Atomic.
2NF: PK simple.
3NF: No transitive dependency.

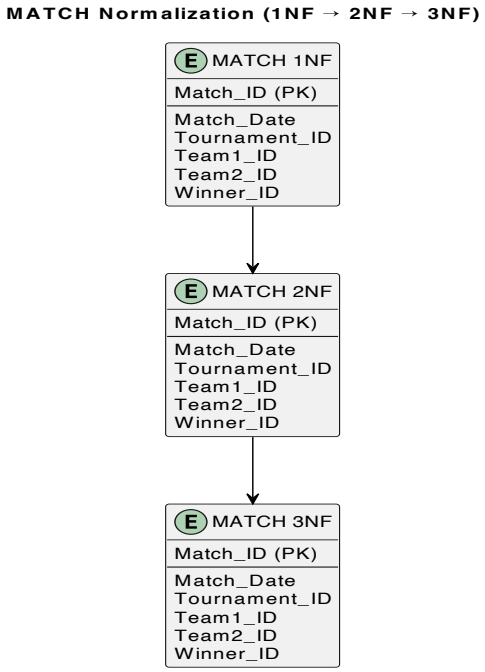


MATCH

1NF: Atomic.

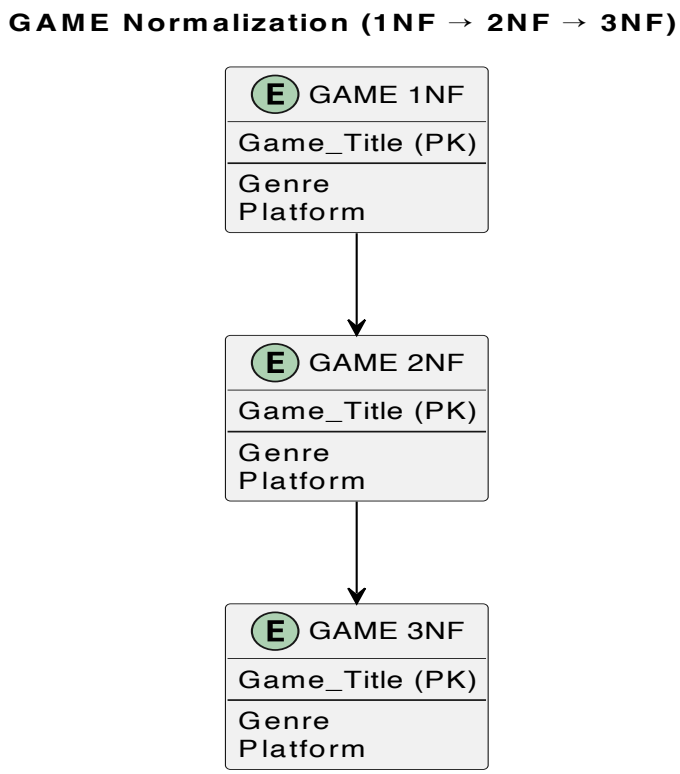
2NF: PK simple.

3NF: No transitive dependency.



GAME

1NF: Atomic.
2NF: PK simple.
3NF: No transitive dependency.

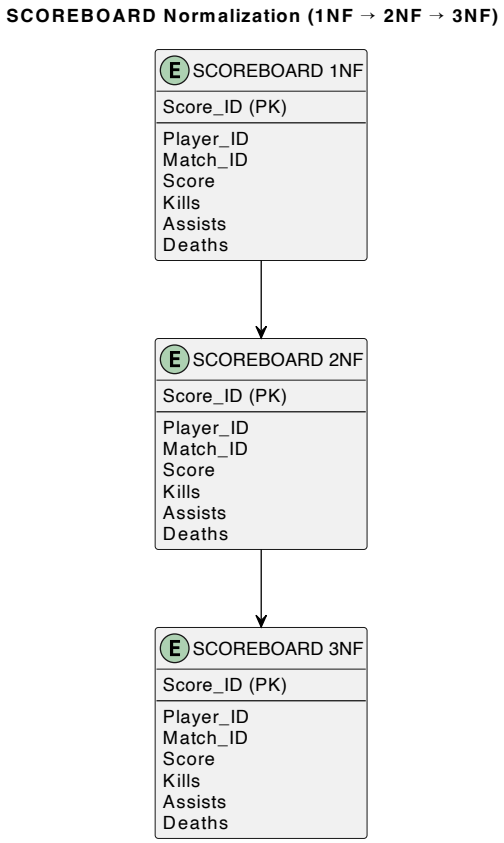


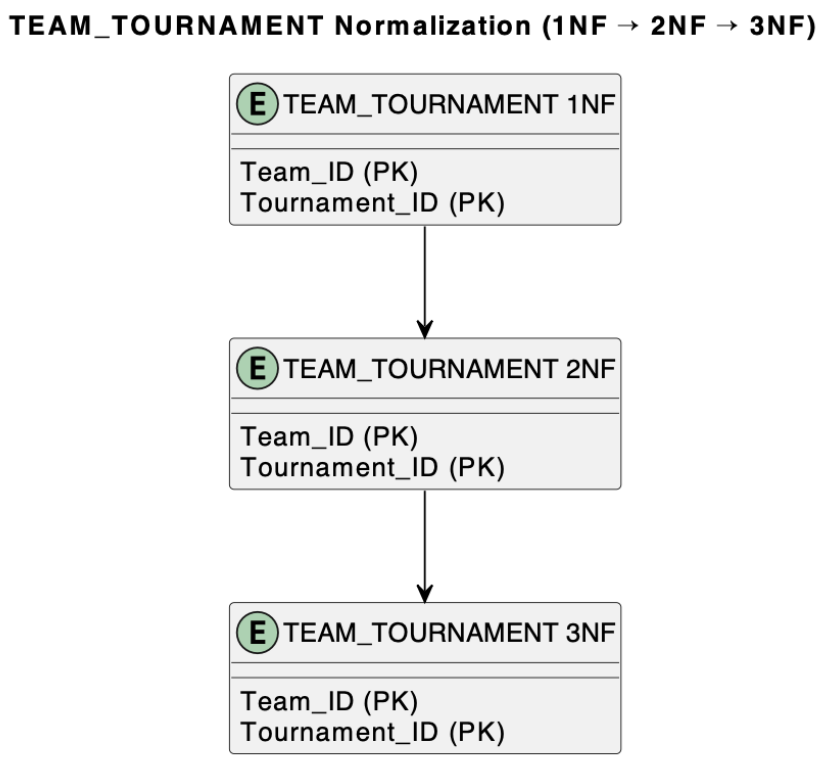
SCOREBOARD

1NF: Atomic.

2NF: PK simple.

3NF: No transitive dependency.



TEAM_TOURNAMENT

1NF: Atomic.
2NF: Composite PK.
3NF: No non-key attributes.